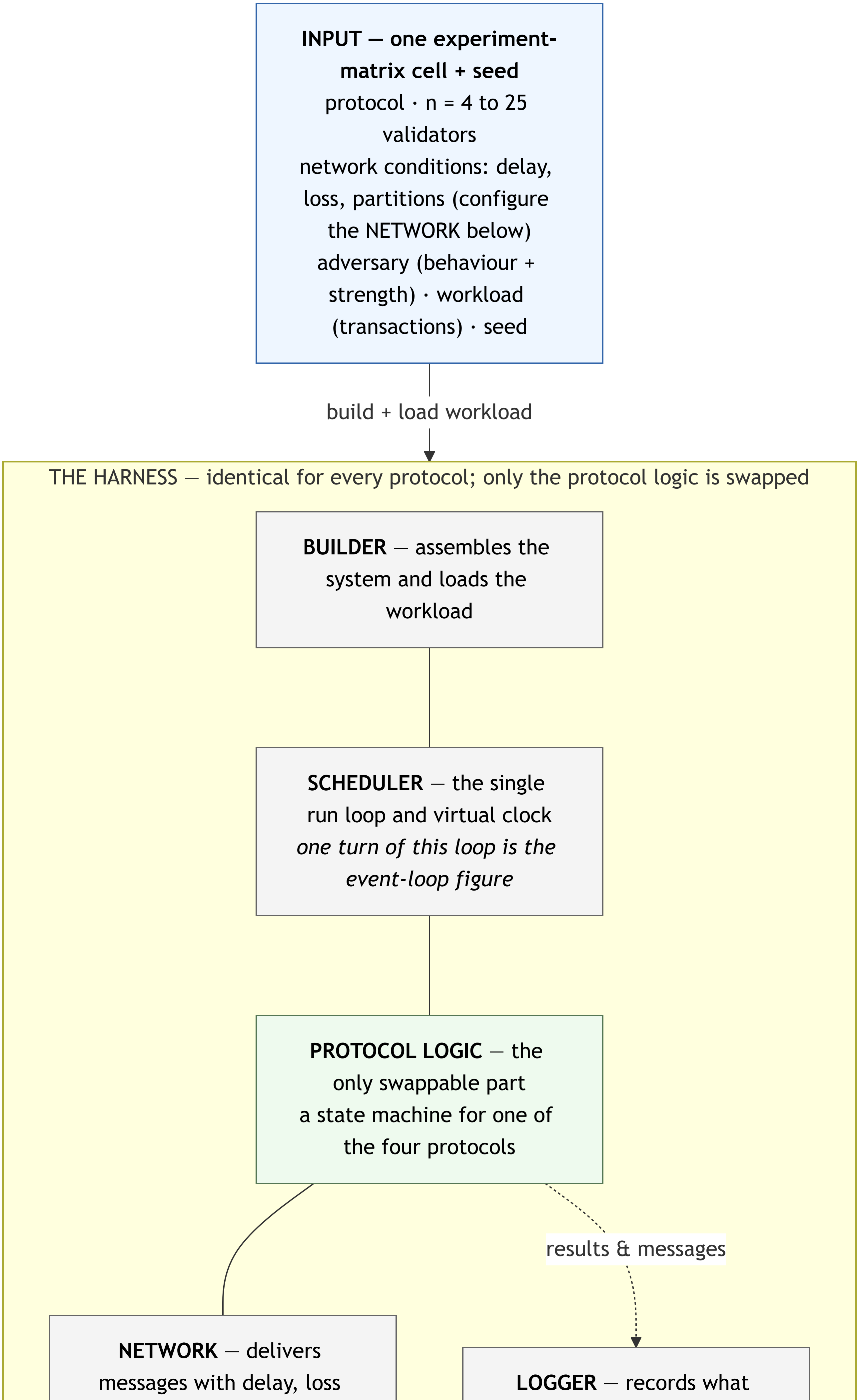
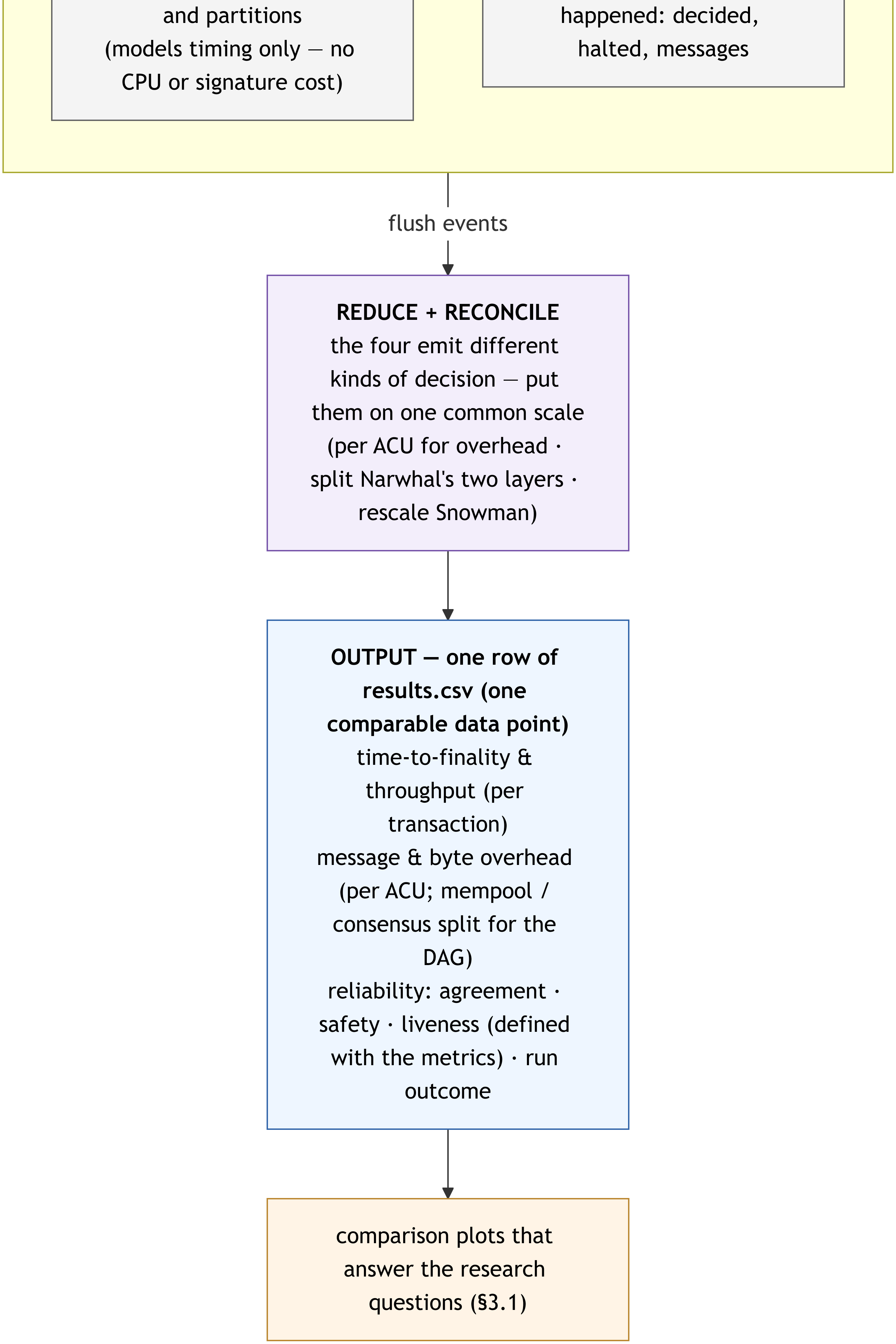

INPUT — one experiment-matrix cell + seed
protocol · n = 4 to 25 validators
network conditions: delay, loss, partitions (configure the NETWORK below)
adversary (behaviour + strength) · workload (transactions) · seed
build + load workload
THE HARNESS — identical for every protocol; only the protocol logic is swapped
BUILDER — assembles the system and loads the workload
SCHEDULER — the single run loop and virtual clock
one turn of this loop is the event-loop figure
PROTOCOL LOGIC — the only swappable part
a state machine for one of the four protocols
results & messages
NETWORK — delivers messages with delay, loss
LOGGER — records what


and partitions (models timing only — no CPU or signature cost)
happened: decided, halted, messages
flush events
REDUCE + RECONCILE
the four emit different kinds of decision — put them on one common scale (per ACU for overhead · split Narwhal's two layers · rescale Snowman)
OUTPUT — one row of results.csv (one comparable data point)
time-to-finality & throughput (per transaction)
message & byte overhead (per ACU; mempool / consensus split for the DAG)
reliability: agreement · safety · liveness (defined with the metrics) · run outcome
comparison plots that answer the research questions (§3.1)